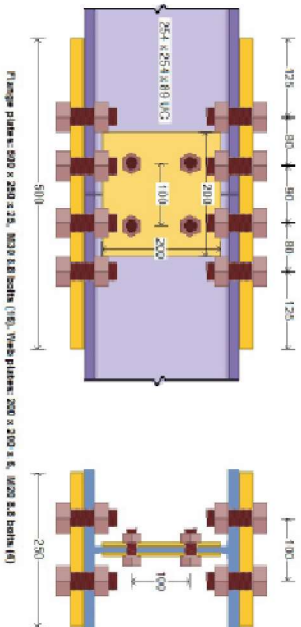


Flange plate splice calculations

Beam: Upper Ground Spine

Beam span: 6.6 m. Section: 254 x 254 x 89 UC S355



Flange plates: 254 x 254 x 89 UC, M20 8.8 bolts (16), Web plates: 200 x 200 x 6, M20 8.8 bolts (6)

Splice 1: 2.4 m. from R1. BM: 185.7 kNm SF: 38.93 kN (factored)

Splice 2: 4.4 m. from R1. BM: 200.0 kNm SF: 26.37 kN (do.)

Section size: 254 x 254 x 89 UC S355

Section dims: D = 260.3 B = 256.3 T = 17.3 t = 10.3 d = 200.3 r = 12.7

Use S275 flange plates 500 x 250 x 25 mm,

Use 2 pairs of M30 8.8 bolts at 100 mm cross centres; inmost pairs of bolts 90 mm apart, 1 further sets of bolts at 60 mm pitch

Web plates: 200 x 200 x 6 mm, 2 pairs of M20 8.8 bolts at 100 mm (H) and 100 mm (V)

Stiffness check: SCI A2043 recommends that flange plate l values should be not less than beam values

Beam/plate I_y : 14,300/25,501 cm⁴; I_y : 4,860/6,510 cm⁴

Basic detailing checks

Flange/Flange Plates: Using M30 bolts, washer diameter 56 mm

Check flange bolt washer clear beam web: pitch (100 mm) $\geq$ 92 mm OKFlange plate bolt edge distance (75.0 mm) $\geq$ 1,25D (37.5 mm) OKFlange bolt edge distance (78.1 mm) $\geq$ 1,25D (37.5 mm) OKPlate plate bolt end distance (126 mm) $\geq$ 1,25D (37.5 mm) OKFlange bolt end distance (45 mm) $\geq$ 1,25D (37.5 mm) OK (n.b. bearing capacity is reduced as $\leq$ 2.0d)Flange bolt spacing (80 mm) $\geq$ 2.5D (75 mm) OK

Web/ Web Plates: Using M20 bolts, washer diameter 37 mm

Check web plate clears beam rods: plate height, 200 mm $\leq$ d, 200.3 mm OKWeb plate bolt end distance H (50 mm) $\geq$ 2.0D (40 mm) OKWeb plate bolt end distance V (50 mm) $\geq$ 2.0D (40 mm) OKWeb plate bolt pitch (100 mm) $\geq$ 2.5D (50 mm) OKWeb bolt end clearance (50 mm) $\geq$ 2.0D (40 mm) OK

All dimensional checks satisfied

Flange plates

Check splice with BM = 200.0 kNm

Axial force in beam flanges = 200.0 x 1000/260.3 = 823 kN

Axial capacities:

Plate: 6,250 x 265/1000 = 1656 kN

Plate net: 1,2 x 4,650 x 265/1000 = 1479 kN

Beam flange: 4,434 x 345/1000 = 1530 kN

Beam flange net: 1,1 x 3,327 x 345/1000 = 1263 kN <<< OK

0.6 x 265 x 25 x 1205 + 1.2 x (175 - 2.5 x 32)/1000 = 1,268 kN OK ($K_g = 1.2$; $k = 2.5$) [2.4]Flange effective area: 3,659 mm² Plate effective area: 5,580 mm²Flange modulus, Z_y : 152,5cm³ Plate modulus, Z_y : 207,1 cm³

Check for strut action moment not required

M30 8.8, net CSA 561 mm², $P_g = 375$ N/mm², $P_{gk} = 460/550$ N/mm² (S275/S355) [Tables 30&32]

Shear resistance per bolt: 581 x 375/1000 = 210 kN

Beam flange bearing resistance per bolt: 30 x 17.3 x 550/1000 = 285 kN

Flange end bolt capacity (reduced as e<2d) = 0.5 x 45 x 17.3 x 550/1000 = 214 kN [2.3.3]

Beam flange bolt group resistance = 2 x 2 x 210 = 842 kN (shear governs)

Flange plate bearing resistance per bolt: 30 x 25 x 460/1000 = 345 kN

Flange plate bolt group resistance = 2 x 2 x 210 = 842 kN (shear governs)

Moment capacity = 842 x 243.0/1000 = 204.5 kNm (limited by bolt group (flange))

Splice moment capacity = 204.5 kNm OK

Web plates

Check splice with SF = 39.93 kN

Shear: (2 plates)

Gross area, $A_{vg} = 2 \times 0.9 \times 200 \times 6 = 2,160$ mm² [2.2.3]Net area, $A_{vn} = 2,160 - (2 \times 2 \times 22 \times 6) = 1,632$ mm²As $A_{vg} \geq 0.85 A_{vn} / K_g$ (1,530, $K_g = 1.2$), gross area, A_{vg} can be used [2.2.3]Shear capacity = 0.6 p_y , $A_{vg} = 0.6 \times 2,160 \times 275/1000 = 356$ kN OK [2.2.3]2 x 0.6 x 265 x 6 x (101 + 1.2 x (100.0 - 0.5 x 32)/1000 = 396 kN OK ($K_g = 1.2$; $k = 0.5$)M20 8.8, net CSA 245 mm², $P_g = 375$ N/mm², $P_{gk} = 460/550$ N/mm² (S275/S355) [Tables 30&32]

Shear capacity per bolt = 2 x 245 x 375/1000 = 184 kN (double shear)

Bearing on web plates per bolt: 2 x 20 x 6 x 460/1000 = 110 kN <<<

Bearing on beam web per bolt: 20 x 10.3 x 550/1000 = 113 kN

Shear on each bolt = 39.93/2 = 19.97 kN

Modulus of web bolt group = $n/n + 1$ p_y = 2 x 3 x 100/6 = 100 mm

Moment load on outermost bolts = 39.93 x 50/100 = 19.97 kN

Load on outermost bolts = $d/(19.97 + 19.97)$ = 28.24 kN OK

Web plate bolts are adequate

Splice shear capacity = 156.1 kN OK

TITLE

PROPOSED DRAWINGS

ADDRESS

95 MALVERN ROAD
LONDON, NW6

CLIENT

Mr. C. NORMAN

SCALE

1:50 @ A3

DATE

MARCH 2019

DRAWING NO.

190328/32

REV.



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